

Results: Language, Topics, and Embeddings

## RQ3: Topical and Linguistic Structure I

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: modafinil, treatment, study, effects, patients, results, sleep, used, use, drug, studies, clinical, mg, using, placebo, cognitive, associated, effect, however, disorder.

**Table 3. NMF topics extracted from the corpus.**

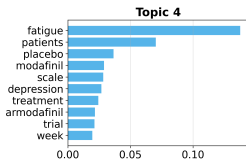
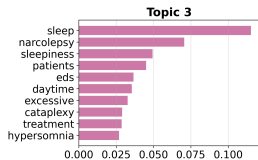
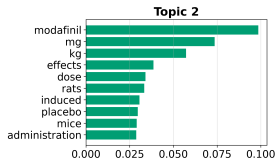
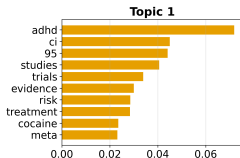
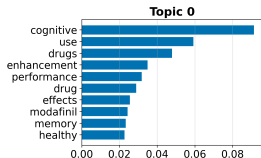
| Topic | Top terms                                                                        |
|-------|----------------------------------------------------------------------------------|
| 0     | cognitive, use, drugs, enhancement, performance, drug, effects, modafinil        |
| 1     | adhd, ci, 95, studies, trials, evidence, risk, treatment                         |
| 2     | modafinil, mg, kg, effects, dose, rats, induced, placebo                         |
| 3     | sleep, narcolepsy, sleepiness, patients, eds, daytime, excessive, cataplexy      |
| 4     | fatigue, patients, placebo, modafinil, scale, depression, treatment, armodafinil |

## RQ3: Topical and Linguistic Structure II

The topics reveal the thematic structure of the literature: Topic 0 centres on cognitive enhancement and neuroenhancement; Topic 1 addresses ADHD treatment and clinical evidence; Topic 2 covers pharmacological dose-response studies (including animal models); Topic 3 focuses on sleep disorders (narcolepsy, excessive daytime sleepiness); and Topic 4 addresses fatigue in psychiatric populations. These topics cross-cut the keyword-based subfield taxonomy, revealing connections that the explicit classification does not capture.

# RQ3: Topical and Linguistic Structure III

**NMF Topic — Top Terms**



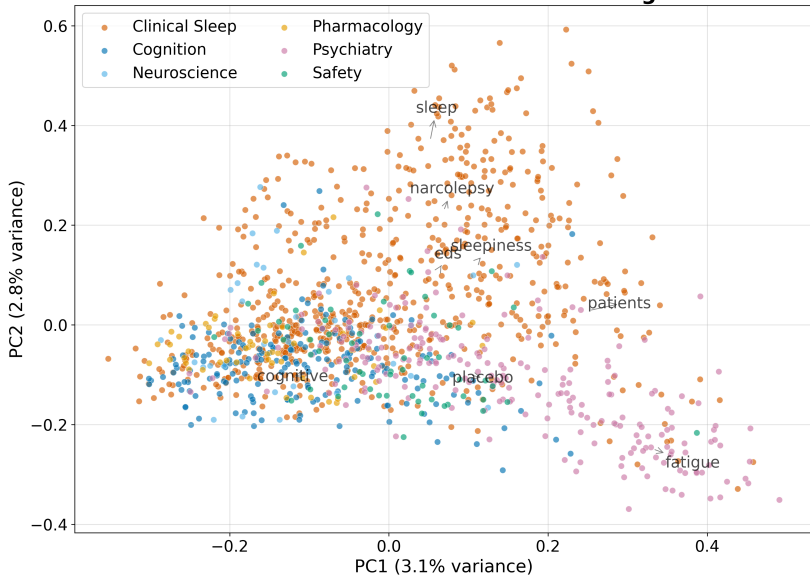
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# Document Embeddings I

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

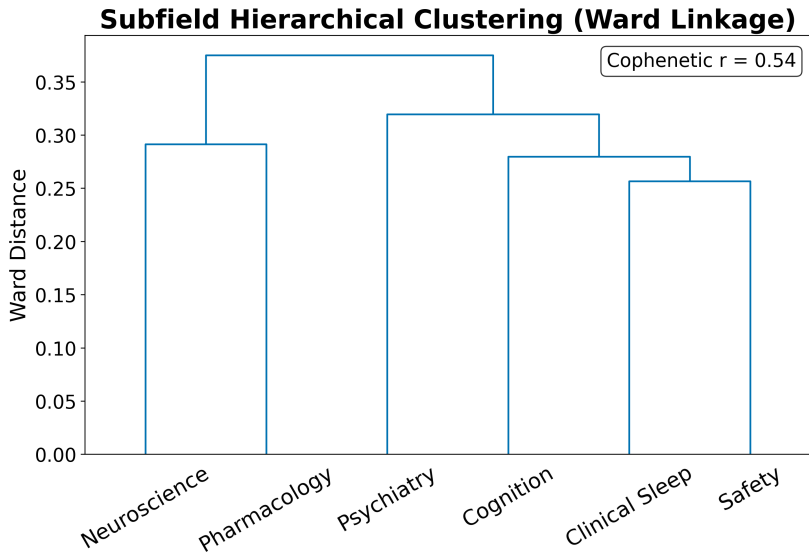
# Document Embeddings II

**PCA of TF-IDF Document Embeddings**



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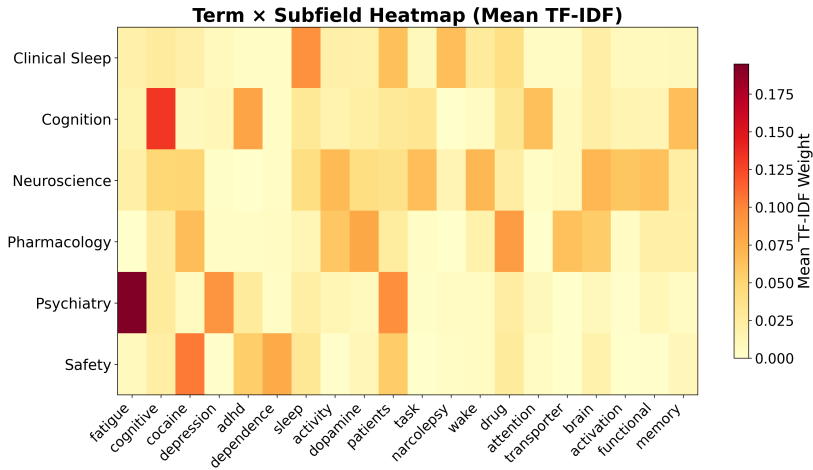
# Document Embeddings III



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# Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.





# Named Entity Analysis I

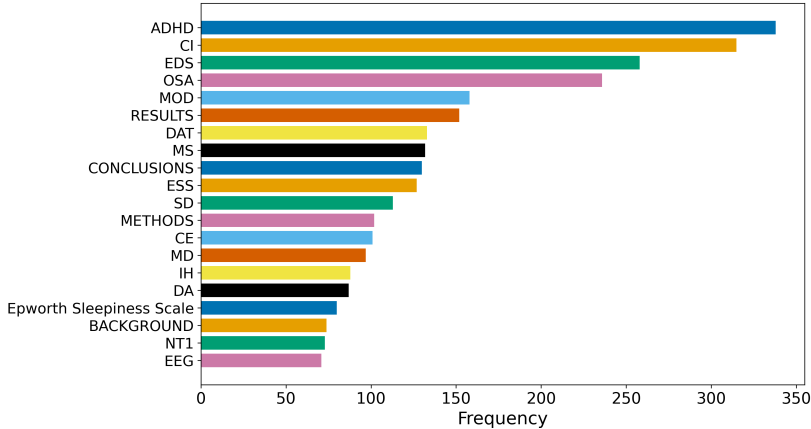
Named entity extraction over the 1277 abstracts identified 30 unique entities. The most frequent entities reflect the clinical and pharmacological vocabulary of the modafinil literature.

**Table 4. Top named entities in abstracts.**

| Entity      | Frequency |
|-------------|-----------|
| ADHD        | 338       |
| CI          | 315       |
| EDS         | 258       |
| OSA         | 236       |
| MOD         | 158       |
| RESULTS     | 152       |
| DAT         | 133       |
| MS          | 132       |
| CONCLUSIONS | 130       |
| ESS         | 127       |
| SD          | 113       |
| METHODS     | 102       |
| CE          | 101       |
| MD          | 97        |
| IH          | 88        |

# Named Entity Analysis II

Top 20 Named Entities in Abstracts



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Table 5. Top keyphrases by TF-IDF score.

| Keyphrase | Score  |
|-----------|--------|
| available | 0.3333 |
| abstract  | 0.3333 |

# Named Entity Analysis III

|                    |        |
|--------------------|--------|
| abstract available | 0.3333 |
| content            | 0.1053 |
| access             | 0.1053 |
| md                 | 0.0870 |
| jama               | 0.0826 |
| cleveland          | 0.0763 |
| modafinil          | 0.0741 |
| depression         | 0.0741 |
| substance          | 0.0694 |
| drug               | 0.0667 |
| conditions         | 0.0667 |
| continuous         | 0.0667 |
| continuous flow    | 0.0667 |

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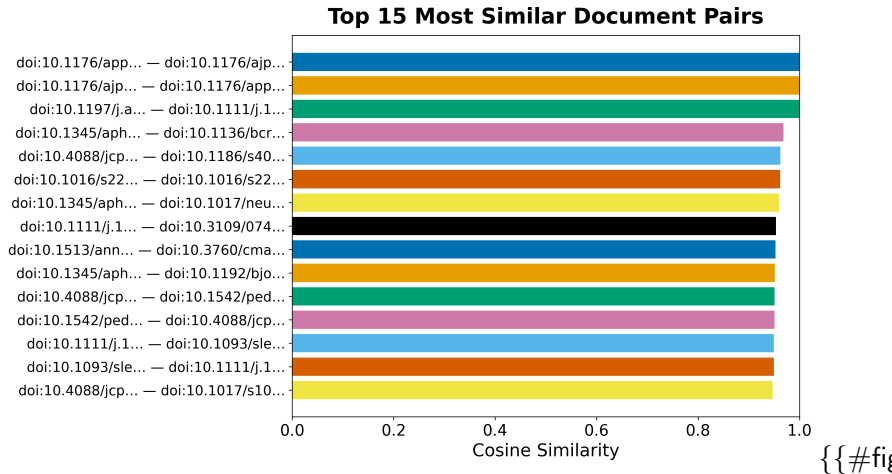
# Embedding Similarity and Clustering I

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with  $k = 5$  clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

**Table 6. Top 10 most similar document pairs.**

| Paper A                        | Paper B                        | Similarity |
|--------------------------------|--------------------------------|------------|
| doi:10.1176/appi.ajp.163.12.21 | doi:10.1176/ajp.2006.163.12.21 | 1.0000     |
| doi:10.1176/ajp.2006.163.12.21 | doi:10.1176/appi.ajp.163.12.21 | 1.0000     |
| doi:10.1197/j.aem.2005.08.013  | doi:10.1111/j.1553-2712.2006.t | 1.0000     |
| doi:10.1345/aph.1h302          | doi:10.1136/bcr.08.2011.4652   | 0.9687     |
| doi:10.4088/jcp.09m05900gry    | doi:10.1186/s40345-015-0034-0  | 0.9628     |
| doi:10.1016/s2215-0366(18)3026 | doi:10.1016/s2215-0366(25)0006 | 0.9621     |
| doi:10.1345/aph.1h302          | doi:10.1017/neu.2023.6         | 0.9598     |
| doi:10.1111/j.1365-2869.2008.0 | doi:10.3109/07420528.2011.6352 | 0.9538     |
| doi:10.1513/annalsats.202006-6 | doi:10.3760/cma.j.cn112147-202 | 0.9532     |
| doi:10.1345/aph.1h302          | doi:10.1192/bjo.2024.75        | 0.9517     |

# Embedding Similarity and Clustering II



# Embedding Similarity and Clustering III

## Literature Corpus — Term Cloud



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## Embedding Similarity, and Clustering IV

